



Lecture 3

3.1 Half-Range Expansions: The Cosine and Sine Series
ハーフレンジ拡張：CosineおよびSine級数

3.2 Mean-Square Approximation and Parseval's Identity
平均二乗近似とパーセヴァルの等式

3.1 Half-Range Expansions: The Cosine and Sine Series

ハーフレンジ拡張：

CosineおよびSine級数

3.1 Half-Range Expansions: The Cosine and Sine Series

- In many applications we are interested in representing by a Fourier series a function $f(x)$ that is defined only in a finite interval, say $0 < x < p$.
- Since this $f(x)$ is clearly not periodic, the Fourier series results of the previous lectures are not readily applicable.
- Our goal in this lecture is to show how we can represent $f(x)$ by a Fourier series, after extending it to a periodic function.

3.1 Half-Range Expansions: The Cosine and Sine Series

Theorem 3.1 Half-Range Expansions

Suppose that $f(x)$ is a piecewise smooth function defined on an interval $0 < x < p$.

Then f has a cosine series expansion

$$a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi}{p} x \quad (0 < x < p) \quad (3.1)$$

where

$$a_0 = \frac{1}{p} \int_0^p f(x) dx \quad \text{and} \quad a_n = \frac{2}{p} \int_0^p f(x) \cos \frac{n\pi}{p} x dx \quad (n \geq 1) \quad (3.2)$$

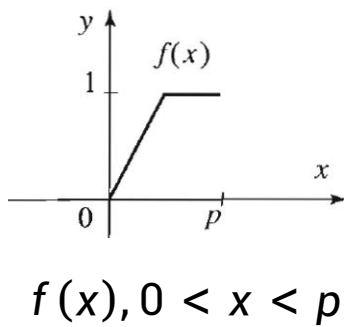
Also, f has a sine series expansion

$$\sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{p} x \quad (0 < x < p) \quad (3.3)$$

where

$$b_n = \frac{2}{p} \int_0^p f(x) \sin \frac{n\pi}{p} x dx \quad (n \geq 1) \quad (3.4)$$

On the interval $0 < x < p$, the series (3.1) and (3.3) converge to $f(x)$ if $f(x)$ is continuous at x and to $\frac{f(x-) + f(x+)}{2}$ otherwise.



3.1 Half-Range Expansions: The Cosine and Sine Series

- The series (3.1) and (3.3) are commonly referred to as **the half-range expansions of f** .
- They are **two different series representations of the same function on the interval $0 < x < p$** .

3.1 Half-Range Expansions: The Cosine and Sine Series

- Define the even periodic extension of f by

$$f_2(x) = \begin{cases} f(x) & \text{if } 0 < x < p \\ f(-x) & \text{if } -p < x < 0 \end{cases}$$

and $f_2(x) = f_2(x + 2p)$ otherwise.

- Define the odd periodic extension of f by

$$f_1(x) = \begin{cases} f(x) & \text{if } 0 < x < p \\ -f(-x) & \text{if } -p < x < 0 \end{cases}$$

and $f_1(x) = f_1(x + 2p)$ otherwise.

Q1: Plot $f_2(x)$

Q2: Plot $f_1(x)$

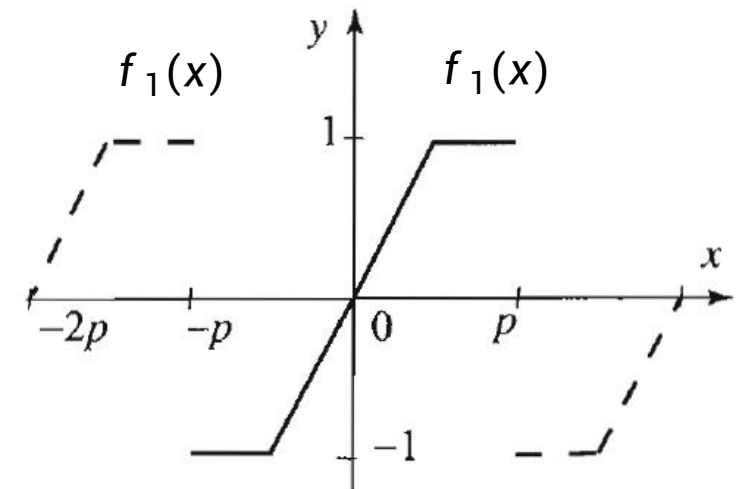
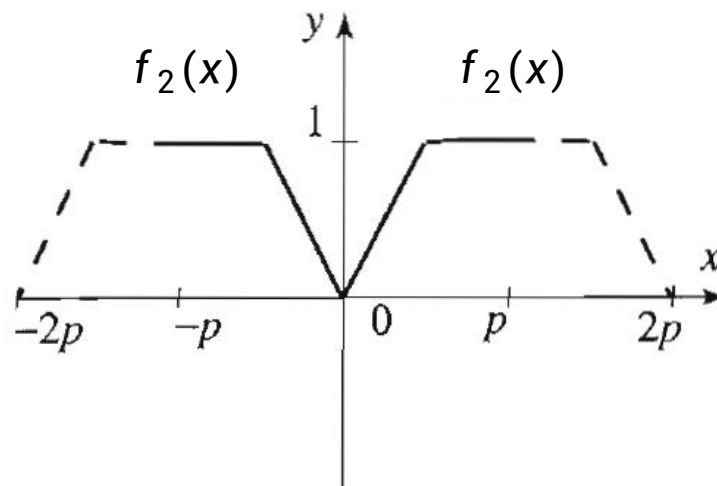
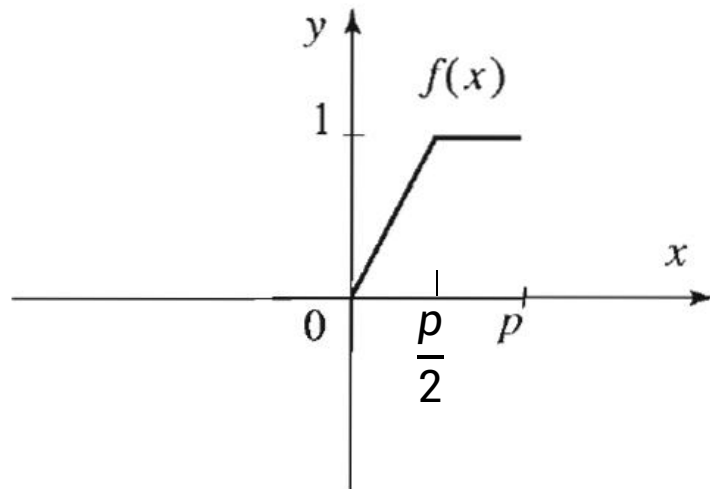


Figure 3.1 (a) $f(x)$, $0 < x < p$

(b) Even $2p$ -periodic extension, $f_2(x)$. (c) Odd $2p$ -periodic extension, $f_1(x)$.

3.1 Half-Range Expansions: The Cosine and Sine Series

Example 3.1 Half-range expansions

Find the half-range expansions of the function $f(x) = x$ for $0 < x < 1$.

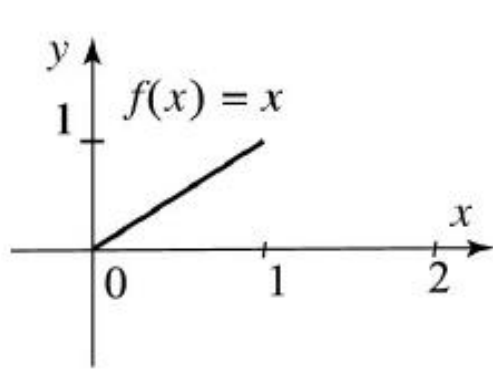
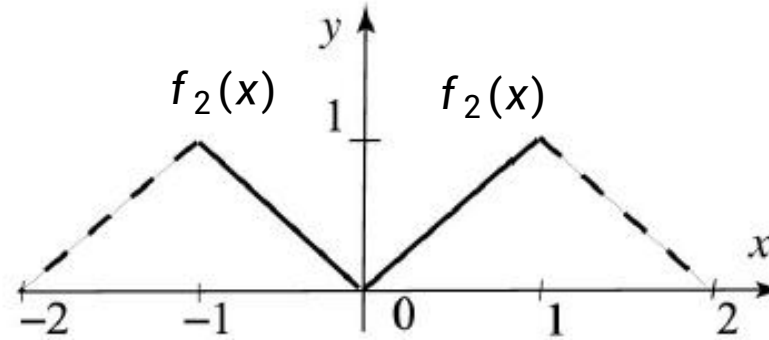
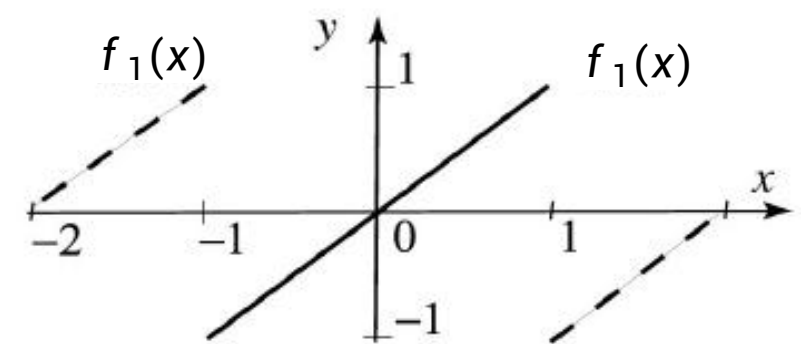


Figure 3.2 (a) $f(x) = x, 0 \leq x \leq 1$



(b) Even extension of f , period 2.

Q1: Plot even extension $f_2(x)$



(c) Odd extension of f , period 2.

Q2: Plot odd extension $f_1(x)$

3.1 Half-Range Expansions: The Cosine and Sine Series

Solution:

(1) We expand $f(x)$ as even function shown in Figure 3.2(b).

It is actually a special case of Example 2.6 of Lecture 2 with $p=1$.

We have

$$\begin{aligned} f_1(x) = x &= \frac{1}{2} - \frac{4}{\pi^2} \left(\cos \pi x + \frac{1}{3^2} \cos 3\pi x + \frac{1}{5^2} \cos 5\pi x + \dots \right) \\ &= \frac{1}{2} - \frac{4}{\pi^2} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} \cos(2k+1)\pi x, \text{ for all } 0 < x < 1 \end{aligned}$$

(2) We expand $f(x)$ as odd function shown in Figure 3.2(c).

By using (3.4) in Theorem 3.1, we see

$$b_n = \frac{2}{p} \int_0^p f(x) \sin \frac{n\pi}{p} x \, dx \quad (n \geq 1)$$

$$= \frac{2}{1} \int_0^1 x \cdot \sin n\pi x \, dx \quad (n \geq 1)$$

$$= -\frac{2 \cos(n\pi)}{n\pi} = \frac{2(-1)^{n+1}}{n\pi}$$

Hence $f(x)$ has its sine series expansion

$$f_2(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{p} x \quad (0 < x < p)$$

$$\text{i.e. } x = \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n\pi} \sin \frac{n\pi}{1} x, \quad 0 < x < 1$$

$$= \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin n\pi x, \quad 0 < x < 1$$

3.1 Half-Range Expansions: The Cosine and Sine Series

Example 3.2 Half-range expansions

Find the half-range expansions of the function $f(x) = \sin x$ for $0 \leq x \leq \pi$.

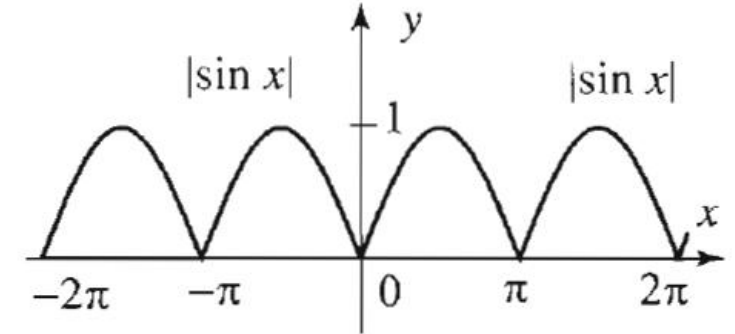
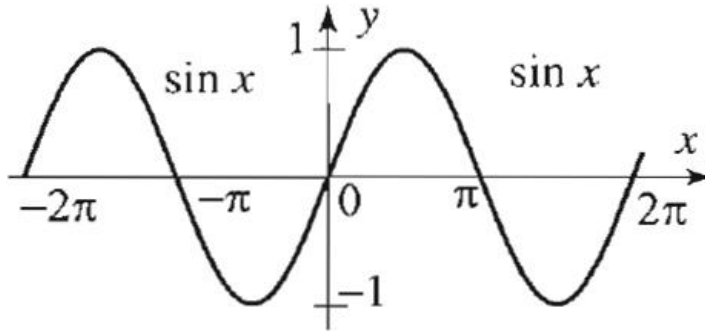
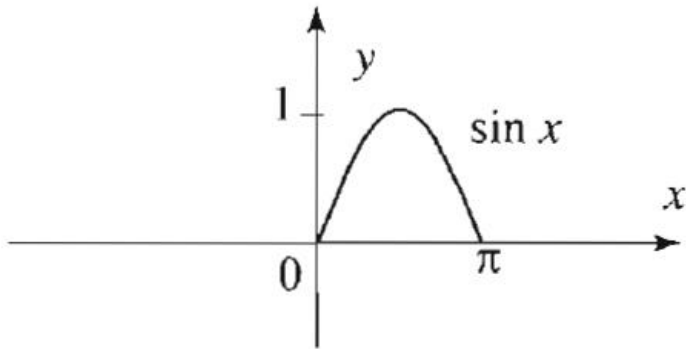


Figure 3.3 (a) $f(x) = \sin x, 0 \leq x \leq \pi$ (b) Odd extension of f , $f_1(x) = \sin x$. (c) Even extension of f , $f_2(x) = |\sin x|$.

Solution

Q2: Plot odd
extension $f_1(x)$

Q1: Plot even
extension $f_2(x)$

(1) Consider the function $f(x) = \sin x, 0 \leq x \leq \pi$. If we take its **odd extension**, we get the usual sine function, **$f_1(x) = \sin x$ for all x** . Thus, the sine series expansion is just $\sin x$.

3.1 Half-Range Expansions: The Cosine and Sine Series

Solution (cont.)

(2) If we take the **even** extension of f , we get the function $|\sin x|$, notice that it is 2π -periodic function.

Using the Theorem 2.3 (Fourier Series of Even and Odd Functions) in Lecture 2, we have

$$a_0 = \frac{1}{\pi} \int_0^{\pi} \sin x \, dx = -\frac{1}{\pi} \cos x \Big|_0^{\pi} = \frac{2}{\pi}$$

For $n \geq 1$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos \frac{n\pi x}{\pi} dx = \frac{2}{\pi} \int_0^{\pi} \sin x \cos nx \, dx = \frac{2}{\pi} \int_0^{\pi} \frac{1}{2} [\sin(1-n)x + \sin(1+n)x] dx$$

Because $\sin a \cos b = \frac{1}{2} (\sin(a-b) + \sin(a+b))$
https://en.wikipedia.org/wiki/List_of_trigonometric_identities

$$= \frac{1}{\pi} \left[-\frac{1}{1-n} \cos(1-n)x - \frac{1}{1+n} \cos(1+n)x \right]_0^{\pi} \quad (\text{if } n \neq 1)$$

$$= \frac{1}{\pi} \left[-\frac{1}{1-n} (-1)^{1-n} - \frac{1}{1+n} (-1)^{1+n} + \frac{1}{1-n} + \frac{1}{1+n} \right]$$

$$= \begin{cases} 0 & \text{if } n \text{ is odd} \\ \frac{4}{\pi(1-n^2)} & \text{if } n \text{ is even} \end{cases}$$

For $n = 1$, we have

Because $\sin 2\theta = 2\sin \theta \cos \theta$

$$a_1 = \frac{2}{\pi} \int_0^{\pi} \sin x \cos x \, dx = \frac{1}{\pi} \int_0^{\pi} \sin(2x) \, dx = 0$$

Thus, the Fourier series is:

$$f_2(x) = |\sin x| = \frac{2}{\pi} - \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{1}{(2k)^2 - 1} \cos 2kx, \quad 0 \leq x \leq \pi$$

3.2 Mean-Square Approximation and Parseval's Identity

平均二乗近似とパーセヴァルの等
式

3.2 Mean-Square Approximation and Parseval's Identity

Series (級数)

Review Calculus II (微積分 II)

Calculate the addition for the terms of an infinite sequence $\{a_n\}_{n=1}^{\infty}$ we get

$$a_1 + a_2 + a_3 + \dots + a_n + \dots$$

which is called an **infinite series** (or just a **series**) and is denoted by

$$\sum_{n=1}^{\infty} a_n \quad \text{or} \quad \sum a_n$$

3.2 Mean-Square Approximation and Parseval's Identity

From

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64} + \cdots + \frac{1}{2^n} + \cdots$$

a_n (pointing to $\frac{1}{2^n}$)

we get

$$\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \frac{15}{16}, \frac{31}{32}, \frac{63}{64}, \dots, 1 - \frac{1}{2^n}, \dots$$

$$\sum a_n$$

n	Sum of first n terms
1	0.50000000
2	0.75000000
3	0.87500000
4	0.93750000
5	0.96875000
6	0.98437500
7	0.99218750
10	0.99902344
15	0.99996948
20	0.99999905
25	0.99999997

We can observe that

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots + \frac{1}{2^n} + \cdots = 1$$

3.2 Mean-Square Approximation and Parseval's Identity

Review Calculus II (微積分 II)

Consider the **partial sums** (部分和)

$$s_1 = a_1$$

$$s_2 = a_1 + a_2$$

$$s_3 = a_1 + a_2 + a_3$$

$$s_4 = a_1 + a_2 + a_3 + a_4$$

and in general

$$s_n = a_1 + a_2 + a_3 + \cdots + a_n = \sum_{i=1}^n a_i$$

3.2 Mean-Square Approximation and Parseval's Identity

When a $2p$ -periodic function is represented by its Fourier series,

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{p} x + b_n \sin \frac{n\pi}{p} x \right)$$

it is important to know how well the N th partial sums approximate f .

$$s_N(x) = a_0 + \sum_{n=1}^N \left(a_n \cos \frac{n\pi}{p} x + b_n \sin \frac{n\pi}{p} x \right) \quad (3.5)$$

Q: How well the partial sums of the Fourier series approximate the function $f(x)$?

To tackle this problem, we will study the integral

$$E_n = \frac{1}{2p} \int_{-p}^p [f(x) - s_N(x)]^2 dx \quad (3.6)$$

known as the **mean (or total) square error** of the partial sum S_N relative to $f(x)$.

We also say that S_N approximates $f(x)$ in the mean with error E_N .

3.2 Mean-Square Approximation and Parseval's Identity

Example 3.3 Approximation in the mean by Fourier series

Compute E_N for $N = 1, 2, \dots, 10$, in the case of the 2π -periodic

sawtooth function $f(x) = \frac{1}{2}(\pi - x)$, $0 < x < 2\pi$

Solution:

The Fourier series was computed in Example 1.3 of Lecture 1

Thus we have $S_N(x) = \sum_{n=1}^N \frac{\sin nx}{n}$.

From (3.b)

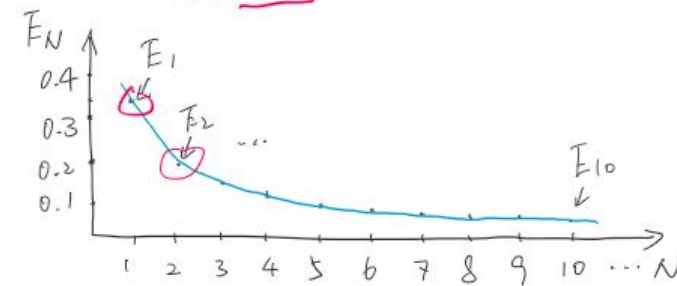
$$\begin{aligned} E_N &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (f(x) - S_N(x))^2 dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left[\frac{1}{2}(\pi - x) - \sum_{n=1}^N \frac{\sin nx}{n} \right]^2 dx \\ &= \frac{1}{2\pi} \int_0^{2\pi} \left[\frac{1}{2}(\pi - x) - \sum_{n=1}^N \frac{\sin nx}{n} \right]^2 dx \end{aligned}$$

Now we try different N , for example, from 1 to 10.

$$E_1 = \frac{1}{2\pi} \int_0^{2\pi} \left[\frac{1}{2}(\pi - x) - \sin x \right]^2 dx = 0.322$$

Then we have approximations of E_N for small N shown in the following table.

N	1	2	3	4	5	6	7	8	9	10
E_N	0.322	0.197	0.142	0.111	0.091	0.077	0.067	0.059	0.053	0.048



The table seems to indicate that E_N decreases to zero as N increases. That is

$$\lim_{N \rightarrow \infty} E_N = \lim_{N \rightarrow \infty} \frac{1}{2\pi} \int_0^{2\pi} (f(x) - S_N(x))^2 dx = 0$$

3.2 Mean-Square Approximation and Parseval's Identity

We introduce the class of **square integrable functions** on

$[a, b]$, which consists of functions f defined on $[a, b]$ and such that

$$\int_{-p}^p f^2(x) dx < \infty$$

Theorem 3.2 Approximation in the Mean by Fourier Series

Suppose that f is **square integrable** on $[-p, p]$. Then S_N , the N th partial sum of the Fourier series of $f(x)$, approximates (or converges to) $f(x)$ in the mean with **an error E_N** that decreases to zero as $N \rightarrow \infty$. In symbols, we have

$$\lim_{N \rightarrow \infty} E_n = \lim_{N \rightarrow \infty} \frac{1}{2p} \int_{-p}^p [f(x) - s_N(x)]^2 dx = 0 \quad (3.7)$$

If we integrate and assume that we can interchange the integral and the limit, we get

$$\lim_{N \rightarrow \infty} \frac{1}{2p} \int_{-p}^p [f(x) - s_N(x)]^2 dx = \frac{1}{2p} \int_{-p}^p \lim_{N \rightarrow \infty} [f(x) - s_N(x)]^2 dx = \frac{1}{2p} \int_{-p}^p 0 dx = 0$$

3.2 Mean-Square Approximation and Parseval's Identity

Theorem 3.3 Mean Square Error

Suppose that $f(x)$ is square integrable on $[-p, p]$. Then

$$E_n = \frac{1}{2p} \int_{-p}^p f^2(x) dx - \left[a_0^2 + \frac{1}{2} \sum_{n=1}^N (a_n^2 + b_n^2) \right] \quad (3.8)$$

Theorem 3.3 allows us to compute the mean square error E_n in terms of the integral of the square of the function and its Fourier coefficients of index less than or equal to N .

3.2 Mean-Square Approximation and Parseval's Identity

Proof:

We expand the right side of (3.6), we get

$$\begin{aligned} E_n &= \frac{1}{2p} \int_{-p}^p (f(x) - S_N(x))^2 dx \\ &= \frac{1}{2p} \int_{-p}^p [f(x)^2 - 2f(x)S_N(x) + S_N(x)^2] dx \\ &= \frac{1}{2p} \int_{-p}^p f(x)^2 dx - \underbrace{\frac{1}{p} \int_{-p}^p f(x)S_N(x) dx}_{(1)} + \underbrace{\frac{1}{2p} \int_{-p}^p S_N(x)^2 dx}_{(2)}. \end{aligned}$$

From the definition of S_N and the Fourier coefficients (2.8), (2.9), (2.10) in Lecture 2.

It follows that

$$\begin{aligned} (1) &= -\frac{1}{p} \int_{-p}^p f(x) S_N(x) dx \\ &= -\frac{1}{p} \int_{-p}^p f(x) \left(a_0 + a_1 \cos \frac{\pi}{p} x + \dots + a_N \cos \frac{N\pi}{p} x \right. \\ &\quad \left. + b_1 \sin \frac{\pi}{p} x + \dots + b_N \sin \frac{N\pi}{p} x \right) dx \\ &= -(2a_0^2 + a_1^2 + \dots + a_N^2 + b_1^2 + \dots + b_N^2). \end{aligned}$$

Expanding $S_N(x)^2$, then integrating term by term while using the orthogonality of the trigonometric system, we obtain

$$(2) = \frac{1}{2p} \int_{-p}^p S_N(x)^2 dx = a_0^2 + \frac{1}{2} a_1^2 + \dots + \frac{1}{2} a_N^2 + \frac{1}{2} b_1^2 + \dots + \frac{1}{2} b_N^2$$

Substituting the values of the last two integrals in the expression for E_n , we obtain (3.8). Q.E.D.

3.2 Mean-Square Approximation and Parseval's Identity

One useful consequence of (3.8) is the following **inequality**:

$$a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \leq \frac{1}{2p} \int_{-p}^p f^2(x) dx$$

known as **Bessel's inequality** (ベッセルの不等式).

To **prove** it, from (3.6), note that $E_N \geq 0$. Hence (3.8) implies that

$$a_0^2 + \frac{1}{2} \sum_{n=1}^N (a_n^2 + b_n^2) \leq \frac{1}{2p} \int_{-p}^p f^2(x) dx$$

The desired inequality follows by letting $N \rightarrow \infty$.

3.2 Mean-Square Approximation and Parseval's Identity

Appealing to Theorem 3.2, and using (3.8), it follows that

$$0 = \lim_{N \rightarrow \infty} E_n = \frac{1}{2p} \int_{-p}^p f^2(x) dx - \left[a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right]$$

or, equivalently,

Parseval's Identity (パーセヴァルの等式)

$$\frac{1}{2p} \int_{-p}^p f^2(x) dx = a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \quad (3.9)$$

3.2 Mean-Square Approximation and Parseval's Identity

It is valid for all **square integrable functions** (自乗可積分函数) on $[-p, p]$ and has many interesting applications.

Parseval's identity, or **Bessel's inequality**, imply that the Fourier coefficients of a **square integrable function** are **square summable**. That is, we have

$$a_0^2 + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) < \infty$$

You should note that, in general, we do not have

$$\sum_{n=1}^{\infty} a_n < \infty \quad \text{or} \quad \sum_{n=1}^{\infty} b_n < \infty$$

3.2 Mean-Square Approximation and Parseval's Identity

Example 3.4 Evaluating series with Parseval's identity

Evaluate the series $1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$ (Hint: Example 3.3)

Solution:

$$\text{We see } 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

From Example 3.3, $S_N(x) = \sum_{n=1}^N \frac{\sin nx}{n}$, we notice its Fourier coefficients

$a_n = 0$ for $n = 0, 1, 2, \dots$ and $b_n = \frac{1}{n}$ for $n = 1, 2, 3, \dots$.

Putting this into Parseval's identity,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)^2 dx = \underbrace{a_0^2}_{=0} + \frac{1}{2} \sum_{n=1}^{\infty} (\underbrace{a_n^2}_{=0} + \underbrace{b_n^2}_{=1/n^2})$$

we get $f(x) = \frac{1}{2}(\pi - x)$

$$\frac{1}{2\pi} \int_0^{2\pi} \left(\frac{1}{2}(\pi - x)\right)^2 dx = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n^2} \quad \textcircled{1}$$

$$\text{Since left } \textcircled{1} = -\frac{1}{24\pi}(\pi - x)^3 \Big|_0^{2\pi} = \frac{\pi^2}{12}$$

We get

$$\frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 2 \cdot \frac{\pi^2}{12} = \frac{\pi^2}{6}$$

Review for Lecture 3

- Half-Range Expansions: The Cosine and Sine Series
- Mean-Square Approximation
- Parseval's Identity

Exercise

Please Check <https://web-ext.u-aizu.ac.jp/~xiangli/teaching/MA05/index.html>

Reading: Lecture 3 Slides

References

- [1] Nakhlé H. Asmar, *Partial Differential Equations with Fourier Series and Boundary Value Problems 2nd Edition*, 2004
- [2] Wikipedia